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Venetoclax combined with intensive chemotherapy as induction chemotherapy in newly diagnosed AML with FLT3-ITD-mutation

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Dear Editor

The FMS-internal tandem duplication (FLT3-ITD) is found in approximately 20–25% of de novo AML. Improvements in overall survival (OS) have been achieved with the development of FLT3 inhibitors (midostaurin, quizartinib, and gilteritinib) and allogeneic hematopoietic stem-cell transplantation (allo-HSCT) [1–3]. Consequently, the 2022 European Leukemia Network (ELN) guidelines reclassified FLT3-ITD-mutated AML as intermediate risk disease. However, the complete remission (CR) rate has not been significantly increased by conventional chemotherapy combined with FLT3 inhibitors [4]. The optimal treatment for patients with FLT3-ITD mutation is still being explored.

Venetoclax (VEN) combined with intensive chemotherapy (IC) increased both CR and minimal residual disease (MRD)-negative rates in patients with de novo AML significantly [4–6]. However, the efficacy of VEN in combination with IC has not been established for FLT3-ITD-mutated AML. In this study, we retrospectively analyzed 63 patients with de novo FLT3-ITD-mutated AML who received VEN (7 days) and IC as induction therapy to verify the efficacy and safety of this combination in patients with FLT3-ITD mutations.

We retrospectively analyzed the clinical data of 63 de novo AML patients who were treated with VEN plus IC as induction therapy from two clinical trials (VEN + DA: ChiCTR2200061524; VEN + HAA: ChiCTR205893472) and one retrospective cohort (VEN + HAD), conducted at 13 hospitals in China between January 2022 and August 2024. The FLT3-ITD mutation with an allelic ratio of ≥ 0.5 was defined as high allele ratio. All inclusion and exclusion criteria were fully described in our previous study [5, 6]. All investigations were conducted with the approval of the institutional review boards and in accordance with the principles of the Declaration of Helsinki. Written informed consent was obtained from all participants before trial enrollment.

Patients received VEN in combination with IC as induction chemotherapy. The IC regimens included: DA (Daunorubicin 60 mg/m² intravenously [IV] daily on days 2–3 plus cytarabine 200 mg/m² IV on days 2–7), HAA (Homoharringtonine 2.5 mg/m² IV daily on days 3–7, cytarabine 100 mg/m² IV daily on days 3–7, and aclacinomycin 20 mg/m² IV daily on days 3–7), HAD (Homoharringtonine 2 mg/m² IV daily on days 2–6, daunorubicin 45 mg/m² IV daily on days 4–5, and cytarabine 100 mg/m² IV daily on days 2–6). VEN was used at a dose escalation model, 100 mg, 200 mg, 400 mg/d for 7-days. Potent CYP3A inhibitors such as voriconazole and posaconazole, which were used for antifungal treatment or prophylaxis, mandated a dose reduction of VEN to

100 mg once daily. Allo-HSCT was recommended for patients after achieving CR or CR with incomplete hematologic recovery (CRi) according to ELN 2022 guidelines. Patients who did not undergo or were ineligible for allo-HSCT received consolidation and maintenance therapy. The treatment process is detailed in our previous study [5, 6].

The primary endpoint was to evaluate cCR (CR+CRi) after one cycle of induction chemotherapy. Secondary endpoints included OS, event-free survival (EFS), relapse-free survival (RFS), and the safety profile of the induction therapy. Early mortality was defined as death within 30 days of initiation of chemotherapy. Categorical variables were compared using Fisher's exact test. Survival outcomes were analyzed using the Kaplan-Meier method, with intergroup comparisons performed using the log-rank test.

Between January 2022 and August 2024, 63 AML patients with de novo FLT3-ITD-mutated were enrolled. The median age was 40 years (range, 16–59), and 30 patients (47.6%) were male. The median allelic ratio of the FLT3-ITD mutation was 0.53 (range: 0.01–14.75), and 59 patients (93.7%) harbored at least one additional mutation besides FLT3-ITD. Gene mutations are depicted in Supplementary Fig. 2. The baseline characteristics of the study population are summarized in Table 1.

Among enrolled patients, 28 received VEN + DA, 26 received VEN + HAA, and 9 received VEN + HAD. One patient died and another withdrew from the study prior to treatment initiation. Of the 61 evaluable patients, the overall response rate (ORR) (CR + partial remission (PR)) was 96.7% (59/61; 95% CI: 87.8–99.2) with the CR rate was 91.8% (56/61; 95% CI: 81.8–96.5), and 88.7% (47/53; 95% CI: 80.1–97.2) of patients achieved MRD-negative CR by multiparameter flow cytometry. According to the 2022 ELN risk stratification, the CR rates were 90.9% (10/11; 95% CI: 56.1–98.7) in the favorable group, 92.3% (24/26; 95% CI: 73.9–98.1) in the intermediate group, and 91.7% (22/24; 95% CI: 72.2–97.9) in the adverse group. Comparison of the different treatment regimens, the CR rates were 96.3% (26/27; 95% CI: 77.9–99.5) in the VEN + DA group, 96.0% (24/25; 95% CI: 76.5–99.4) in the VEN + HAA group, and 66.7% (6/9; 95% CI: 35.9–97.5) in the VEN + HAD group ($p = 0.037$; Suppl. Table 2). Five patients who did not achieve CR received salvage treatment. Out of the two patients with no response (NR), one received sorafenib combined with VEN plus azacitidine, while the other was treated with sorafenib combined with IC; both achieved CR. One of the three PR patients who were treated with sorafenib in combination with IC achieved CR.

To November 30, 2025, with a median follow-up of 21 months (range, 1–42), 13 patients underwent HSCT, consisting of 12

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Table 1. Baseline characteristics of the 63 AML patients.

Characteristic	Entire cohort	VEN + DA	VEN + HAA	VEN + HAD	<i>p</i>
Total number of patients	63	28(44.4%)	26(41.3%)	9(14.3%)	
Median age, years (range)	40(16–59)	47(16–59)	42(18–59)	33(23–58)	0.025
Gender					0.616
Male	30(47.6%)	15(53.6%)	12(46.2%)	6(66.7%)	
Female	33(52.4%)	13(46.4%)	14(53.8%)	3(33.3%)	
FLT3-ITD allelic ratio	0.53(0.01–14.75)				0.912
High (> 0.5)	28(44.4%)	11(39.3%)	10(38.5%)	4(44.5%)	
Low (< 0.5)	26(41.3%)	12(42.9%)	12(46.2%)	5(55.6%)	
Unknown	9(14.3%)	5(17.9%)	4(15.4%)	0(0%)	
ECOG	1(0–2)	1(0–2)	1(0–2)	1(0–2)	0.998
Baseline WBC ($\times 10^9/L$) Median (IQR)	35.42(7.0–82.7)	25.23(7.25–62.75)	45.57(11.15–127.94)	13.74(2.66–85.03)	1.000
Baseline platelet($\times 10^9/L$) Median (IQR)	41(24–69)	32.5(16.5–50.5)	43.0(33.8–78.8)	50.0(31.5–168.5)	0.617
ELN 2022 risk group					0.523
Favorable	11(17.5%)	7(25.0%)	3(11.5%)	1(11.1%)	
Intermediate	27(42.8%)	9(32.1%)	14(53.8%)	4(44.4%)	
Adverse	25(39.7%)	12(42.9%)	9(34.6%)	4(44.4%)	

ECOG Eastern Cooperative Oncology Group performance status. Data are presented as median (range) for continuous variables and n (%) for categorical variables. IQR interquartile range, WBC white blood cell count.

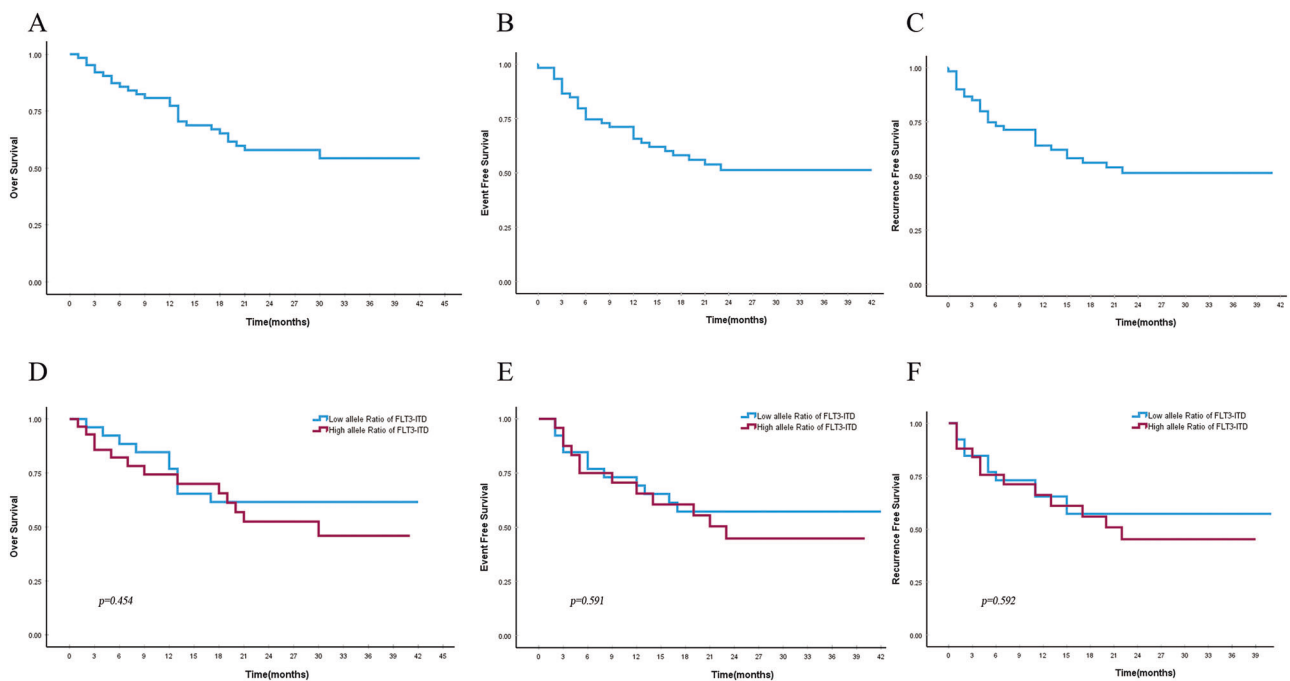


Fig. 1 The Kaplan-Meier method was used to estimate overall survival (OS), event-free survival (EFS), and disease-free survival (RFS) for all enrolled AML patients (A–C); comparisons of OS, EFS, and RFS between the low and high FLT3-ITD allelic ratio subgroups are also presented (D–F).

(19.0%, 12/63) allo-HSCT and 1 (1.6%, 1/63) autologous-HSCT. Disease recurrence was observed in 24 (38.1%, 24/63) patients, and 26 (41.3%, 26/63) patients died by the end of follow-up. The median OS, EFS, and RFS were not reached in the entire cohort. The estimated 2-year OS, EFS, and RFS rates were 57.9% (95% CI: 45.2–70.6), 51.3% (95% CI: 37.8–64.8), and 51.4% (95% CI: 37.9–64.9) for the entire cohort, respectively (Fig. 1A–C). No significant survival differences were observed between low and high FLT3-ITD allelic ratio groups (Fig. 1D–F), across 2022 ELN risk strata, or among the three treatment subgroups (Suppl. Fig. 3).

Grade 3 or worse adverse events, including neutropenia, thrombocytopenia, and febrile neutropenia, occurred in all patients during induction treatment. The 30-day mortality rate was 1.6%.

Standard induction chemotherapy yields CR rates of only 40–60% among patients with FLT3-ITD-mutated AML [7–10]. Although addition of FLT3 inhibitors has improved survival outcomes, it has not translated into meaningful improvements in remission rates in most clinical studies [1–3, 11]. In our study, patients with FLT3-ITD mutations who received VEN in

combination IC achieved CR rate of 91.8%, ORR of 96.7%, and MRD-negative CR rate of 88.7% after one cycle of induction chemotherapy. Subgroup analysis demonstrated that the CR rate was over 90% in all ELN risk groups. The therapeutic efficacy was significantly superior to that reported in previous studies using IC combined with FLT3 inhibitor regimens [12, 13]. This study offers a novel reference for treatment options in newly diagnosed AML patients with FLT3-ITD mutations.

To November 30, 2025, with a median follow-up of 21 months (range, 1–42), 12 (19.0%) patients underwent allo-HSCT. The median OS, EFS, and RFS were not reached in the entire cohort. The estimated 2-year OS, EFS, and RFS were 57.9%, 51.3%, and 51.4%, respectively. No significant differences were observed by ELN risk, treatment subgroups, or FLT3-ITD allelic ratio. Despite the development of novel therapies, allo-HSCT remains the preferred treatment modality according to both the ELN and Chinese guidelines and is a critical determinant of long-term survival for patients with FLT3-ITD-mutated AML [14]. In the RATIF trial, in which 59% patients underwent allo-HSCT, the 4-year OS and EFS rates were 51.4% and 28.2%, respectively, in the midostaurin group [12]. Pratz KW et al. showed that with a 72% allo-HSCT rate in FLT3-ITD-mutated patients, the OS rate at 4 years was 59% [3]. In our study, only 19.0% of patients underwent allo-HSCT for multiple reasons, and the 2-year OS was 57.9%, which is similar to previous studies. The EFS (51.3%) was significantly improved compared with that in the RATIF trial [12]. This result is encouraging. It may benefit from: First, VEN combined with IC achieved higher CR and MRD negative rates in FLT3-ITD-mutated AML patients than FLT3-ITD inhibitors combined with IC. This enables patients to achieve a deeper remission during induction therapy, which is critical for long-term survival. Second, our regimen systematically applied VEN throughout the whole treatment course (induction, consolidation, and maintenance), which may reduce the relapse rate and prolong OS. In addition, optimization of chemotherapy doses (reducing chemotherapy doses during induction and consolidation) improves treatment tolerance and patient compliance. Regarding safety, the combination regimen was well tolerated with no significant increase in treatment-related toxicity due to limiting venetoclax to a 7-day course and dose adjustment of intensive chemotherapy [4].

This study has several limitations. First, the sample size was relatively small and did not allow for further detailed analysis. Second, the follow-up time was too short to draw a conclusion on long-term survival benefits. Third, as a single-arm study design, there was no control group for comparison. In future studies, we will continue to expand the number of patients, extend the follow-up time, and conduct randomized controlled trials to further verify the efficacy of VEN combined with IC in the treatment of FLT3-ITD mutated de novo AML patients.

In recent years, triple-targeted therapy (Ven combined with azacitidine and gilteritinib) has shown significant efficacy in treating FLT3-mutated acute myeloid leukemia (AML), achieving a CR rate of 90% in de novo AML with FLT3-ITD mutations [15]. The therapeutic efficacy is similar to the results we reported. However, for AML patients who are eligible for intensive chemotherapy, there is a notable absence of randomized controlled trials that compare the effectiveness of this triple-targeted approach with the combination of Ven and intensive chemotherapy as induction regimens.

In conclusion, VEN in combination with intensive chemotherapy demonstrated highly efficient and safe induction regimen for de novo FLT3-ITD-mutated AML patients. It provides a new therapeutic option for these patients.

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DATA AVAILABILITY

The datasets used during the current study are available from the corresponding author on reasonable request. We encourage investigators interested in data sharing and collaboration to contact the corresponding author.

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AUTHOR CONTRIBUTIONS

Yingchang Mi, Kaiqi Liu, Zeping Zhou designed the study. Yuyu Liu, Guanchen Bai, Xingli Zhao, Xiaohui Suo, Yanliang Bai, Fang Zheng were participated in therapeutic regimen discussion. Yuyu Liu and Kaiqi Liu collected the data, and wrote the manuscript. Yuyu Liu, Guanchen Bai, Xingli Zhao, Xiaohui Suo, Yanliang Bai, Fang Zheng, Weihua Zhao, Hongling Peng, Dongmei Wang, Liyun Zhao, Jie Liu, Ling Li, Zeping Zhou, Yingchang Mi, Kaiqi Liu were involved in patient management, patient follow-up, and clinical data collection. Ling Li collated the data and analyzed it. Yingchang Mi and Kaiqi Liu revised the manuscript and provided valuable advice. All authors read and approved the final manuscript. ALL authors had full access to all the data and had final responsibility for the decision to submit for publication.

COMPETING INTERESTS

The authors declare no competing interests.

ETHICS APPROVAL AND CONSENT TO PARTICIPATE

The study protocol was conducted in accordance with the principles of the Declaration of Helsinki (as revised in 2013) and approved by the Ethics Committee of The affiliated Taian City Central Hospital of Qingdao University, Taian, Shandong, China. Written informed consent was obtained from the parents/guardians.

CONSENT FOR PUBLICATION

All authors have read and approved the manuscript for publication.

ADDITIONAL INFORMATION

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